

Trellis Positioning Source Document

Status: Working draft for internal and agency use

Purpose: This document contains the strategic source material from which Trellis marketing and sales communications can be developed. It is not intended to be used verbatim as marketing copy.

Items marked **[OPEN]** require validation or a decision from the Trellis team.

1. Who is Trellis for?

Primary market

Trellis is for organisations developing biological processes where experiments are expensive, time-consuming and limited—and where teams must make consequential development decisions from incomplete evidence.

The initial markets are:

- Biopharmaceutical process development
- Industrial biotechnology and fermentation process development

Primary customer

The primary customer is the person accountable for process-development progress and outcomes.

Typical roles include:

- Head or Director of Process Development
- Head or Director of Upstream Process Development
- VP of Process Development
- Head of Process Sciences
- Fermentation or Bioprocess Development Lead

They are responsible for ensuring that programmes reach a productive, robust and transferable process within the available time, material and experimental capacity.

Senior sponsor

Depending on the organisation, the economic buyer or senior sponsor may be:

- Chief Technology Officer
- Chief Scientific Officer

- VP of CMC
- Head of R&D
- Head of Process Sciences

These leaders care about development timelines, portfolio capacity, technical risk, manufacturing readiness and the quality of decisions being made across programmes.

Primary users

The people who would work most directly with Trellis include:

- Process-development scientists
- Upstream-development scientists
- Fermentation scientists
- Principal scientists
- Bioprocess engineers
- Modelling and data-science specialists

Other stakeholders

Trellis may also create value for:

- Scale-up and technology-transfer teams
- Strain- and cell-line-development teams
- Programme leaders
- Manufacturing-sciences teams
- Digital and data teams

Recommended initial focus

Trellis should initially speak most directly to the **Head or Director of Process Development**.

This person is close enough to the scientific work to understand the problem, senior enough to own the outcome and motivated to improve how the entire team learns from experiments.

The first beachhead market:

- Biopharmaceutical upstream process development

2. What progress are they responsible for?

Process-development leaders are responsible for turning a biological system into a process that is:

- Productive

- Robust
- Reproducible
- Sufficiently understood
- Economically viable
- Ready for scale-up and transfer

Their responsibility is not simply to run experiments or identify a promising condition. They must progressively reduce uncertainty and make decisions such as:

- Which variables matter most?
- Which interactions require investigation?
- What should the team test next?
- Is the programme learning quickly enough?
- Should the team persist with the current direction?
- Is a change in strategy required?
- When is the evidence sufficient to move forward?
- Where is the process likely to remain robust?
- What trade-offs exist between yield, quality, cost, time and operability?

The underlying progress they are trying to create is:

Moving from limited observations to reliable process understanding, and from process understanding to confident development decisions.

3. What prevents that progress?

Core problem

The hard part of bioprocess development is not simply finding a promising condition.

It is learning a complex biological process from a small number of expensive experiments—and knowing what to test next.

Scientific and operational pain points

Biological processes have large experimental spaces and interacting variables. Teams cannot test every possible combination.

As a result:

- Experiments are expensive and slow.
- Experimental capacity and biological material are limited.
- Variables interact in ways that are difficult to predict.
- Useful signals may be hidden by noise and biological variability.
- Important effects may only appear under particular combinations of conditions.
- Teams must make decisions before they have complete information.

- It is often unclear which next experiment would be most informative.
- A locally promising result may not represent a robust operating region.

The workflow itself adds friction:

- Data is spread across instruments, spreadsheets, notebooks and systems.
- Scientists spend significant time cleaning and restructuring data.
- Experimental design, analysis and modelling often happen in separate tools.
- Analysis frequently takes place only after an experimental campaign is complete.
- Statistical and modelling methods may require specialist support.
- Models are difficult to update consistently as new evidence arrives.
- Assumptions, rejected ideas and decision rationale are easily lost between rounds.

Management pain points

For development leaders, these scientific and workflow problems create broader management challenges:

- Limited visibility into whether a programme is genuinely progressing
- Difficulty distinguishing activity from meaningful learning
- Uncertainty about how to allocate scarce laboratory capacity
- Dependence on a small number of modelling or data specialists
- Inconsistent decision quality across scientists and programmes
- Difficulty reviewing why particular experiments or decisions were chosen
- Loss of knowledge during handovers and personnel changes
- Weak continuity between experimental campaigns

Business consequences

When teams do not learn efficiently from experiments, the consequences may include:

- More experimental cycles
- Higher development costs
- Longer development timelines
- Delayed programme milestones
- Late discovery of weak process directions
- Costly process redesign
- Fragile or underperforming processes
- Delays in scale-up or technology transfer
- Reduced portfolio capacity
- Greater uncertainty around manufacturing readiness and time to market

4. Why do current approaches fall short?

Bioprocess-development teams already use many capable tools and methods, including:

- Design-of-experiments software
- Spreadsheets
- Statistical-analysis packages
- Machine-learning notebooks
- Mechanistic modelling software
- Electronic laboratory notebooks
- Laboratory-information systems
- Data platforms
- Generic AI assistants
- Internal experts and consultants

The problem is not that these tools have no value. The problem is that they usually address isolated parts of the development workflow.

The integration gap

Experimental questions, designs, data, assumptions, models, uncertainty and decisions are often handled in different places.

Between experimental rounds, teams must manually reconstruct:

- What they were trying to learn
- What evidence was generated
- What assumptions were made
- Which models were considered
- What remains uncertain
- Why a particular next experiment was selected

The scientist becomes the manual integration layer between disconnected tools.

This creates three important limitations:

1. **The learning process is fragmented.**
Each experimental round risks becoming a separate analysis exercise rather than part of a cumulative programme of learning.
2. **Models are often retrospective.**
Models may explain completed experiments but are not consistently used to decide what evidence should be generated next.
3. **Scientific context is lost.**
Data may be preserved while the reasoning, assumptions and decision history around the data are not.

Position on existing approaches

Trellis should not claim that established methods such as DoE, mechanistic modelling or machine learning are inadequate.

Trellis should position them as valuable methods that become more useful when they are connected within a continuous, scientist-governed learning process.

5. What category does Trellis belong to?

Recommended category

Model-driven bioprocess development

Product descriptor

Trellis is a scientist-led platform for model-driven bioprocess development.

What “model-driven” means

Model-driven development does not mean that a model makes decisions autonomously.

It means that models are actively used to:

- Structure what the team knows
- Expose what remains uncertain
- Compare possible explanations
- Identify important variables and interactions
- Explore operating regions and trade-offs
- Select informative next experiments
- Update process understanding as evidence accumulates

Scientists define the development question, review the evidence and retain decision authority.

What Trellis is not

Trellis is not:

- A generic AI assistant
- A black-box optimisation engine
- An autonomous scientist
- A replacement for scientific judgement
- A standalone DoE package
- A standalone mechanistic-modelling tool
- Merely a repository for experimental data

6. What is the Trellis value proposition?

Canonical value proposition

Trellis is a scientist-led platform for model-driven bioprocess development. It connects experimental design, process data, modelling and next-experiment selection so that each experimental round sharpens process understanding and makes the next decision clearer.

Short version

Trellis helps bioprocess-development teams learn more from limited experiments and decide what to test next.

Market idea

Every experiment should make the next decision clearer.

Problem–solution statement

Bioprocess teams must understand complex systems using a limited number of expensive experiments. Their tools support individual tasks, but the learning process across experimental rounds remains fragmented.

Trellis connects that process. It helps scientists design experiments, integrate evidence, build and compare models, understand uncertainty and identify useful next experiments within one continuous learning loop.

Outcome statement

Trellis is intended to help teams reach robust process understanding with fewer low-information cycles and greater confidence in the decisions made along the way.

[EVIDENCE GAP] Claims relating to reductions in experiments, timelines or costs to be added once case study by Anmol is completed.

7. What is distinctive about the Trellis approach?

The continuous learning loop

Trellis organises development around a repeated learning process:

1. **Frame**
Define the development objective, response variables, constraints, existing knowledge and unresolved questions.
2. **Design**
Select experiments that are appropriate for the current question and expected to produce useful evidence.

3. **Integrate**
Bring experimental results and relevant process context into a consistent analytical structure.
4. **Model**
Apply suitable statistical, mechanistic, machine-learning or hybrid models.
5. **Optimise and decide**
Explore sensitivities, trade-offs, uncertainty and candidate operating regions.
6. **Learn again**
Use the updated understanding to select the next useful experiment or development decision.

Each cycle becomes part of an accumulating body of process knowledge rather than a disconnected analysis.

Scientist-led AI

Trellis uses AI to coordinate scientific work, preserve context and help scientists interact with specialised methods.

The AI does not replace the underlying scientific and analytical methods. Defined statistical, engineering and modelling tools perform the analysis.

The scientist:

- Defines the question
- Reviews assumptions
- Selects appropriate methods
- Interprets the evidence
- Approves experiments
- Retains decision authority

Multiple modelling approaches

No single model is appropriate for every biological process or every stage of development.

Trellis is intended to support and connect:

- Statistical models
- Mechanistic models
- Machine-learning models
- Hybrid models

This allows teams to use the model appropriate to the available evidence, scientific understanding and development question.

Sequential, uncertainty-aware learning

Trellis treats process development as a sequence of decisions rather than a single optimisation exercise.

The objective is not only to predict the best apparent condition. It is also to understand:

- What the available evidence supports
- Where the model is uncertain
- Which alternatives remain plausible
- Which next experiment would resolve a consequential uncertainty

Preserved scientific context

Trellis aims to keep the following elements connected across experimental rounds:

- Development questions
- Experimental designs
- Data
- Assumptions
- Models
- Uncertainty
- Interpretations
- Decisions
- Decision rationale

This preserved context helps process knowledge become cumulative and reviewable.

8. What is the Trellis USP?

Recommended USP

Trellis keeps the experimental question, evidence, models, uncertainty and decisions connected across development rounds—and uses the evolving process model to help scientists identify the next useful experiment.

Supporting differentiators

A connected learning loop

Trellis connects experimental design, analysis, modelling and next-experiment selection instead of treating them as separate activities.

Process knowledge that compounds

Each round updates the team's understanding rather than requiring the analytical context to be reconstructed.

More than one modelling philosophy

Teams can combine statistical, mechanistic, machine-learning and hybrid approaches rather than being restricted to a single method.

Decisions informed by uncertainty

Trellis is intended to help teams understand not only what a model predicts, but how strongly the available evidence supports the prediction and what should be learned next.

Scientists remain in control

Trellis supports scientific judgement and makes reasoning reviewable. It does not obscure consequential decisions behind autonomous AI.

Built for bioprocess development

The platform is designed around biological process-development questions, constraints and workflows rather than generic analytics.

[CAPABILITY GAP] Each differentiator must be mapped against:

- Available in the current product
- Available with Trellis expert support
- Available in a pilot
- Planned or on the roadmap

Only current and demonstrable capabilities should be presented as existing product features.

9. What benefits does Trellis provide?

The benefits should be understood as a hierarchy. Workflow improvements enable better scientific learning, which can then improve programme and business outcomes.

Immediate workflow benefits

Trellis is intended to help teams:

- Move more quickly from a development question to an experimental design
- Reduce manual data cleaning and restructuring
- Shorten the time between experimental results and the next decision
- Make advanced modelling methods easier to apply
- Compare alternative models more systematically
- Update models as new evidence becomes available
- Preserve assumptions and decision rationale
- Reduce repeated coordination between scientists, analysts and modelling specialists

Scientific benefits

Trellis is intended to help scientists:

- Extract more learning from limited experiments
- Identify important variables and interactions
- Distinguish supported conclusions from uncertain extrapolations
- Understand model sensitivity and uncertainty
- Compare competing explanations of process behaviour
- Identify informative next experiments
- Explore trade-offs between multiple objectives
- Identify candidate operating regions rather than isolated optimum points
- Build process understanding progressively across rounds

Programme benefits

If the workflow and scientific benefits are realised, development teams may be able to:

- Reduce low-information experimental cycles
- Converge on viable process directions sooner
- Detect weak directions earlier
- Improve visibility into programme progress and remaining uncertainty
- Use scarce laboratory capacity more deliberately
- Reduce dependence on a small number of specialists
- Improve consistency in development decisions
- Make handovers and technical reviews more effective

Business benefits

The intended downstream business value includes:

- Shorter development timelines
- Lower experimental burden
- Greater programme capacity
- Reduced risk of late process redesign
- Better-prepared scale-up and technology transfer
- More confidence in manufacturing-readiness decisions
- Better use of scientific and laboratory resources

[EVIDENCE GAP] Programme and business benefits are currently strategic hypotheses unless supported by validated customer or pilot evidence. Marketing should distinguish between demonstrated workflow improvements and anticipated downstream outcomes.

10. Why should customers believe Trellis?

Trellis must support its positioning with evidence rather than broad claims about AI or optimisation. We need to create this section using the case study developed by Anmol.

Product proof

Prospective customers should be able to see:

- How a scientific question is framed
- How existing evidence is incorporated
- How experiments are proposed or designed
- How models are created or compared
- How uncertainty is communicated
- How the next experiment or decision is supported
- How the reasoning remains reviewable by the scientist

Scientific proof

Strong scientific evidence would include:

- Application to a relevant bioprocess dataset
- Comparison against established analysis approaches
- Held-out or prospective validation
- Evidence that the selected experiment improved a consequential decision
- Clear documentation of assumptions and limitations

Workflow proof

Useful operational evidence could include:

- Time required to prepare an experimental design
- Time from receiving data to producing an analysis
- Reduction in manual data preparation
- Reduction in specialist hand-offs
- Ability to update an analysis when new data arrives
- Completeness of the recorded decision trail

Customer proof

Commercial proof could include:

- Paid pilot engagements
- Repeat or expanded engagements
- Customer testimonials
- Case studies
- Research or development partnerships
- Evidence that scientists incorporated Trellis recommendations into actual decisions

Team credibility

Trellis should communicate the team's combined experience across:

- Bioprocess development

- Statistical and mechanistic modelling
- Machine learning and AI
- Scientific software
- Wet-lab experimentation
- Process engineering

[EVIDENCE GAP] The marketing-approved proof set has not yet been documented.
Required inputs include:

- Validated case studies
- Approved customer or partner names
- Quantified workflow results
- Prospective or held-out validation results
- Team credentials and biographies
- Permission to publish specific results

No performance statistic should be invented or published without supporting evidence.

11. What should the customer do next?

Primary call to action

Bring one live development decision.

The strongest entry point is a real, bounded process-development question where:

- The decision matters
- Existing evidence is available
- Important uncertainty remains
- Further experimentation is possible
- Success can be defined in advance

Recommended first engagement

The initial conversation should determine:

- What decision the team needs to make
- What evidence already exists
- What experiments are practical
- What uncertainty is blocking progress
- What success would look like

Where there is a strong fit, the next step should be a bounded paid pilot focused on:

- One process
- One unresolved development decision

- A defined set of evidence
- Agreed success measures
- A clear review point

Potential pilot success measures include:

- Time saved in design or analysis
- Quality of the proposed next experiment
- Reduction in decision-relevant uncertainty
- Scientific usefulness of the model
- Ability to update the model with new results
- Quality and traceability of the decision rationale

[OPEN] Confirm:

- Standard pilot duration
- Pilot deliverables
- Pricing or commercial structure
- Data and experimental requirements
- Customer responsibilities
- Success criteria
- Route from pilot to ongoing deployment

A working assumption is an **8–12 week pilot**, subject to the experimental cadence of the customer's process. This should not be communicated until validated operationally.

12. Positioning summary

One-sentence positioning

Trellis is a scientist-led platform for model-driven bioprocess development that connects experimental design, process data, modelling and next-experiment selection so every experimental round makes the next decision clearer.

Core customer

Heads and Directors of Process Development responsible for delivering productive, robust and transferable biological processes.

Core problem

Teams must learn complex biological processes from a limited number of expensive experiments, while their tools and scientific context remain fragmented across development rounds.

Core promise

Trellis helps scientists extract more useful learning from each experimental round and decide what to test next.

Core distinction

Trellis connects the full learning loop—question, design, evidence, models, uncertainty and decisions—while keeping scientists in control.

Core market idea

Every experiment should make the next decision clearer.

13. Decisions and evidence still required

Before the positioning is treated as final, the Trellis team should resolve the following:

1. **Beachhead market**
Decide whether the initial focus is biopharmaceutical upstream development, industrial biotechnology or a narrower segment.
2. **Precise ideal customer profile**
Define organisation type, company stage, programme stage, team size, process type and buying trigger.
3. **Category language**
Approve “model-driven bioprocess development” or select an alternative.
4. **Capability boundaries**
Document what the product can currently perform independently, what requires expert support and what remains on the roadmap.
5. **Proof**
Approve case studies, results, customer references and measurable claims.
6. **Deployment and trust**
Define the approved position on data security, intellectual property, model governance, deployment and integration.
7. **Pilot offer**
Confirm duration, scope, deliverables, pricing, responsibilities and success measures.
8. **Quality-related scope**
Clarify which product-quality and CMC use cases Trellis currently supports, particularly for mammalian bioprocess development.
9. **External product language**
Decide whether internal names such as KiTTEn or Evidence Engine should appear in customer-facing communication.

Sticky
quality

Get your
audience to...

Tactics

S	Simple	Grasp the "core"	- Prioritization - Schemas - Generative analogies
U	UNEXPECTED! "Wow!" "what!?"	Pay attention, remain interested	- Break a pattern - Use uncommon sense - The gap theory
C	CONCRETE 	Understand & remember	- Images and objects - Experiences - Schemas
C	 credible 	Believe	- Authorities & anti-authorities - Super detailed - Human scale principle - Testable credentials
E	Emotional 	Care	- Personal identity - Self-interest - Maslow's Penthouse
S	Stories "cases" "events"	Act	- Simulation exercises - Inspirational plots -- Challenge plots -- Connection plots -- Creativity plots

~ MORE ~